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Momentum, Impulse & Collisions

Draw before and after. Choose a positive direction. Then only two equations: momentum is conserved, and impulse is change in momentum.

THE TOPIC ON ONE MAP

Momentum, Impulse & Collisions

01 - Momentum and impulse on one body

$$I = mv - mu$$

rebounds

km h⁻¹

02 - Collisions

before = after

direction reversed

03 - One common velocity

coalesce

string taut

ejected

04 - Impulse in a collision

equal and opposite

two possible values

05 - With SUVAT, friction, slopes

dropped / bounces

μ

inclined plane

THE TWO EQUATIONS OF THE WHOLE TOPIC

$$I = mv - mu \quad (= Ft)$$

impulse on **one** particle

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

conservation for the **whole** collision

01 Momentum and Impulse on One Body

Momentum = mv . Impulse = change in momentum = force $\times$ time. Units: **N s**.

WHEN YOU SEE $\rightarrow$ YOU DO

“momentum of ...”

$\rightarrow mv$ — units N s (or kg m s⁻¹)

“impulse given by the force” / “ F applied for t seconds”

$\rightarrow I = Ft = mv - mu$

“rebounds” / “bounces”

$\rightarrow$ the velocity changes sign

km h⁻¹ or grams

$\rightarrow \div 3.6 \cdot \div 1000$ first

1

Choose a positive direction

Draw an arrow. Every velocity the other way is **negative**.

2

One particle only

$I = m(v - u)$ with signs. The sign of I is its direction.

WORKED MODEL • THE BALL REBOUNDS

Ball 0.075 kg hits a ceiling at 5 m s⁻¹ and rebounds at 3 m s⁻¹. Find the impulse.

1 positive = downwards (after): $u = -5$, $v = +3$

signs

2 $I = 0.075(3 - (-5)) = 0.075 \times 8$

$I = m(v - u)$

ANSWER $I = 0.6$ N s

× **TRAP** $0.075(3 - 5)$ is the classic mistake. A rebound **adds** the speeds.

IDEA 01

Practise: Momentum and Impulse on One Body

Questions written to practise exactly this idea.

Q1 DR ESLAM Momentum and units

M1 · TOPIC 9 2 marks

A car of mass 1200 kg moves at 54 km h^{-1} . Find its momentum.

Change km h^{-1} to m s^{-1} first: $\div 3.6$.

Q2 DR ESLAM Rebound: the speeds add

M1 · TOPIC 9 2 marks

A ball of mass 0.2 kg hits a wall at 6 m s^{-1} and rebounds at 4 m s^{-1} . Find the impulse on the ball.

Q3 DR ESLAM Impulse = force $\times$ time

M1 · TOPIC 9 2 marks

A force of 8 N acts for 1.5 s on a 2 kg particle moving at 5 m s^{-1} in the direction of the force. Find its new speed.

02 Collisions: Conservation of Momentum

Total momentum before = total momentum after. Draw it — the diagram gives you the signs.

WHEN YOU SEE → YOU DO

“collide directly”

→ before/after diagram, $\sum mu = \sum mv$

“moving towards each other”

→ one velocity is negative before

“direction of motion is reversed”

→ its sign changes after

“direction unchanged”

→ same sign as before

“state the direction of motion of ...”

→ look at the **sign** of your answer



WORKED MODEL • JUNE 2023

$$1 \quad 4(2u) + 2(-3u) = 4v_A + 2(2u)$$

$$\sum mu = \sum mv$$

$$2 \quad 2u = 4v_A + 4u \Rightarrow v_A = -\frac{1}{2}u$$

solve

ANSWER speed $\frac{1}{2}u$, A is reversed (negative)

× **TRAP** Draw every “after” arrow in the positive direction and let the sign decide. Guessing directions is where the marks are lost.

IDEA 02

Practise: Collisions: Conservation

My questions first, then real past papers.

Q4 DR ESLAM Draw, then conserve

M1 · TOPIC 9 3 marks

A (4 kg) moves at 6 m s^{-1} towards B (2 kg) moving at 3 m s^{-1} . After the collision A moves at 1 m s^{-1} in its original direction. Find the velocity of B.

Positive = A's direction. B starts at -3 .

Q5 DR ESLAM Let the sign decide

M1 · TOPIC 9 3 marks

P (2 kg) at 5 m s^{-1} hits Q (3 kg) at rest. After, Q moves at 4 m s^{-1} . Find the velocity of P and state its direction.

PAST PAPERS

Q6 PAST PAPER Pearson IAL WME01 · January 2022 · Q2

M1 · TOPIC 9 7 marks

2. A particle P has mass km and a particle Q has mass m . The particles are moving towards each other in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P has speed $3u$ and Q has speed u .

As a result of the collision, the direction of motion of each particle is reversed and the speed of each particle is halved.

- (a) Find the value of k .

(4)

- (b) Find, in terms of m and u , the magnitude of the impulse exerted on Q in the collision.

(3)

Q7**PAST PAPER**

Pearson IAL WME01 · January 2023 · Q2

M1 · TOPIC 9**8 marks**

2. Two particles, A and B , are moving in a straight line in opposite directions towards each other on a smooth horizontal surface when they collide directly.

Particle A has mass $3m$ kg and particle B has mass m kg.

Immediately before the collision, both particles have a speed of 1.5 m s^{-1}

Immediately after the collision, the direction of motion of A is unchanged and the difference between the speed of A and speed of B is 1 m s^{-1}

(a) Find (i) the speed of A immediately after the collision,

(ii) the speed of B immediately after the collision.

(5)

(b) Find, in terms of m , the magnitude of the impulse exerted on B in the collision.

(3)

Q8 PAST PAPER Pearson IAL WME01 • June 2023 • Q1

M1 • TOPIC 9 7 marks

1. A particle A has mass 4 kg and a particle B has mass 2 kg.

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of A is $2u \text{ m s}^{-1}$ and the speed of B is $3u \text{ m s}^{-1}$

Immediately after the collision, the speed of B is $2u \text{ m s}^{-1}$

The direction of motion of B is reversed by the collision.

- (a) Find, in terms of u , the speed of A immediately after the collision. (3)
- (b) State the direction of motion of A immediately after the collision. (1)
- (c) Find, in terms of u , the magnitude of the impulse received by B in the collision.
State the units of your answer. (3)

Q9**PAST PAPER**

Pearson IAL WME01 · January 2024 · Q2

M1 · TOPIC 9 6 marks

2.

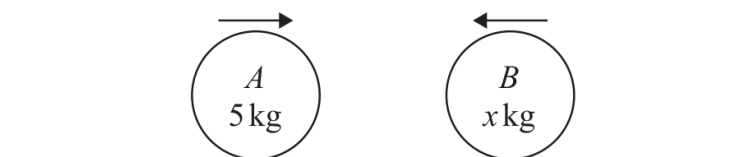
**Figure 2**

Figure 2 shows two particles, A and B , moving in opposite directions on a smooth horizontal surface. Particle A has mass 5 kg and particle B has mass $x\text{ kg}$.

The particles collide directly.

Immediately before the collision, the speed of A is 3 ms^{-1} and the speed of B is $x\text{ ms}^{-1}$

Immediately after the collision, the speed of A is 1 ms^{-1} and its direction of motion is unchanged.

Immediately after the collision, the speed of B is 1.5 ms^{-1}

(a) Find the value of x .

(3)

(b) Find the magnitude of the impulse exerted on A by B in the collision.

(3)

Q10**PAST PAPER**

Pearson IAL WME01 · October 2024 · Q1

M1 · TOPIC 9**6 marks**

1. Particle A has mass $4m$ and particle B has mass $3m$.

The particles are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

Immediately **before** the collision, the speed of A is $2x$ and the speed of B is x .

Immediately **after** the collision, the speed of A is y and the speed of B is $5y$.

The direction of motion of each particle is reversed as a result of the collision.

(a) Show that $y = \frac{5}{11}x$. (3)

(b) Find, in terms of m and x , the magnitude of the impulse received by A in the collision. (3)

03 One Common Velocity: Coalesce, Couple, String Taut

When the particles move **together** after, there is only one unknown: the common velocity.

WHEN YOU SEE → YOU DO

“coalesce” / “join together” / “couple together”

→ one mass $m_1 + m_2$, one velocity v

“string becomes taut”

→ common speed — same equation as coalescing

“jumps off” / “is ejected” / “recoils”

→ started together: $(m_1 + m_2)u = m_1v_1 + m_2v_2$

1

Momentum

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

2

Impulse

Take **one** particle: $I = m(v - u)$.

WORKED MODEL • JUNE 2026

Truck A ($5m$) at u , truck B ($3m$) at $1.5u$ towards each other; they couple.

1 $5m(u) + 3m(-1.5u) = 8m v$

one common velocity

2 $0.5mu = 8mv \Rightarrow v = \frac{u}{16}$

solve

3 $I_B = 3m\left(\frac{u}{16} - (-1.5u)\right)$

impulse on B only

ANSWER $v = \frac{u}{16}, I = \frac{75}{16}mu$

IDEA 03

Practise: One Common Velocity

My questions first, then real past papers.

Q11 DR ESLAM Coalesce

M1 · TOPIC 9 2 marks

A (3 kg) at 4 m s^{-1} hits B (2 kg) at rest and they coalesce. Find the common speed.

Q12 DR ESLAM String becomes taut

M1 · TOPIC 9 3 marks

A (m) and B ($2m$) are joined by a slack string. A is projected away from B at $3u$. Find the common speed when the string becomes taut, and the impulse on B.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · October 2019 · Q1

M1 · TOPIC 9 6 marks

1. Two particles, P and Q , have masses $3m$ and $2m$ respectively. The particles are connected by a light inextensible string. Initially P and Q are at rest on a smooth horizontal plane with the string slack.

Particle P is then projected along the plane directly away from Q with speed $4u$. At the same instant, particle Q is projected along the plane in the opposite direction with speed $3u$.

Find

- (a) the common speed of the particles immediately after the string becomes taut, (3)
- (b) the magnitude of the impulse exerted on Q at the instant when the string becomes taut. (3)

Q14**PAST PAPER**

Pearson IAL WME01 · October 2022 · Q1

M1 · TOPIC 9 5 marks

1. A railway truck S of mass 20 tonnes is moving along a straight horizontal track when it collides with another railway truck T of mass 30 tonnes which is at rest. Immediately before the collision the speed of S is 4 m s^{-1}

As a result of the collision, the two railway trucks join together.

Find

- (a) the common speed of the railway trucks immediately after the collision,

(2)

- (b) the magnitude of the impulse exerted on S in the collision, stating the units of your answer.

(3)

Q15**PAST PAPER**

Pearson IAL WME01 · June 2024 · Q1

M1 · TOPIC 9 5 marks

1. Two particles, A and B , have masses m and $3m$ respectively. The particles are connected by a light inextensible string. Initially A and B are at rest on a smooth horizontal plane with the string slack.

Particle A is then projected along the plane away from B with speed U .

Given that the common speed of the particles immediately after the string becomes taut is S

- (a) find S in terms of U .

(2)

- (b) Find, in terms of m and U , the magnitude of the impulse exerted on A immediately after the string becomes taut.

(3)

Q16**PAST PAPER**

Pearson IAL WME01 · June 2026 · Q1

M1 · TOPIC 9 6 marks

1. Railway truck A has mass $5m$ and railway truck B has mass $3m$. The trucks are moving towards each other in opposite directions along the same straight track when they collide directly. Immediately before the collision the speed of A is u and the speed of B is $1.5u$

As a result of the collision, the trucks couple together.

- (a) Find, in terms of u , the common speed of the trucks immediately after the collision.

(3)

- (b) Find, in terms of m and u , the magnitude of the impulse exerted on B by A in the collision.

(3)

(Total for Question 1 is 6 marks)

04 Impulse in a Collision: Equal and Opposite

In a collision A gives B an impulse, and B gives A the same impulse the other way. Use it instead of conservation when the impulse is given.

WHEN YOU SEE → YOU DO

“the impulse exerted on Q by P is ...”

→ use it on Q : $I = m_Q(v - u)$

“impulse exerted on P by Q ”

→ same size, opposite sign

“find the two possible values”

→ the speed after is $\pm$: solve twice

“hence show that $m_2 > \frac{3}{4}m_1$ ”

→ the speed you found must be > 0

1

Use the impulse on the particle it acts on

Positive direction = the direction the impulse pushes that particle.

2

The other particle

Same impulse, opposite sign — or use conservation.

WORKED MODEL • JUNE 2021

P ($3m$) at ku , Q ($5m$) at $2u$ towards each other. After: P speed u , Q speed $3u$ reversed.

$$1 \quad I_Q = 5m(3u - (-2u)) = 25mu$$

impulse on Q

$$2 \quad 3m(v_P - ku) = -25mu \Rightarrow v_P = ku - \frac{25}{3}u$$

opposite on P

$$3 \quad v_P = \pm u$$

speed u : two cases

ANSWER $k = \frac{28}{3}$ or $k = \frac{22}{3}$

IDEA 04

Practise: Impulse in a Collision

My questions first, then real past papers.

Q17 **DR ESLAM** Use the impulse given

M1 · TOPIC 9 **3 marks**

P (2 kg) moves at 5 m s^{-1} . In a collision the impulse on P has magnitude 6 N s. Find the two possible speeds of P after the collision.

$$2v - 10 = \pm 6.$$

PAST PAPERS

Q18 PAST PAPER Pearson IAL WME01 · January 2020 · Q1

M1 · TOPIC 9 8 marks

1. Two particles, P and Q , of mass m_1 and m_2 respectively, are moving on a smooth horizontal plane. The particles are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision, both particles are moving with speed u .

The direction of motion of each particle is reversed by the collision.

Immediately after the collision, the speed of Q is $\frac{1}{3}u$.

- (a) Find, in terms of m_2 and u , the magnitude of the impulse exerted by P on Q in the collision. (3)
- (b) Find, in terms of m_1 , m_2 and u , the speed of P immediately after the collision. (3)
- (c) Hence show that $m_2 > \frac{3}{4}m_1$ (2)

Q19**PAST PAPER**

Pearson IAL WME01 · January 2021 · Q2

M1 · TOPIC 9**6 marks**

2. Two particles, P and Q , have masses $2m$ and m respectively. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal plane. The particles collide directly.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $2u$.

The magnitude of the impulse exerted on Q by P in the collision is $5mu$.

Find

- (a) the speed of P immediately after the collision, (3)
- (b) the speed of Q immediately after the collision. (3)

Q20 PAST PAPER Pearson IAL WME01 • June 2021 • Q1

M1 • TOPIC 9 7 marks

1. A particle P has mass $3m$ and a particle Q has mass $5m$. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately before the collision the speed of P is ku , where k is a constant, and the speed of Q is $2u$.

Immediately after the collision the speed of P is u and the speed of Q is $3u$.

The direction of motion of Q is reversed by the collision.

- (a) Find, in terms of m and u , the magnitude of the impulse exerted on Q by P in the collision. (2)

- (b) Find the two possible values of k . (5)

Q21 PAST PAPER Pearson IAL WME01 · June 2022 · Q1

M1 · TOPIC 9 4 marks

1. Two particles, P and Q , are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision the speed of Q is $2u$. The mass of Q is $3m$ and the magnitude of the impulse exerted by P on Q in the collision is $4mu$.

Find

(a) the speed of Q immediately after the collision, (3)

(b) the direction of motion of Q immediately after the collision. (1)

Q22 PAST PAPER Pearson IAL WME01 • June 2025 • Q1

M1 • TOPIC 9 6 marks

1.

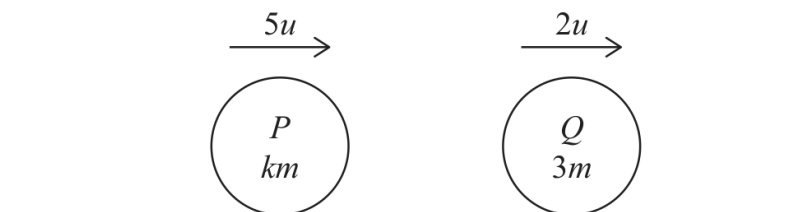
**Figure 1**

Figure 1 shows two particles, P and Q , moving in the same direction along the same straight line on a smooth horizontal surface.

Particle P has mass km and particle Q has mass $3m$

The particles collide directly.

Immediately before the collision, the speed of P is $5u$ and the speed of Q is $2u$

Immediately after the collision, the speed of P is $2u$ and its direction of motion is unchanged.

Immediately after the collision, the speed of Q is v

The impulse exerted on Q by P in the collision has magnitude $4.5mu$

(a) Find v in terms of u only.

(3)

(b) Find the value of k

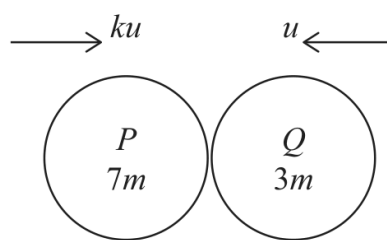
(3)

Q23**PAST PAPER**

Pearson IAL WME01 · October 2025 · Q2

M1 · TOPIC 9 8 marks

2.

**Figure 2**

Particle P of mass $7m$ and particle Q of mass $3m$ are moving in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately **before** the collision, the speed of P is ku and the speed of Q is u , as shown in Figure 2.

Immediately **after** the collision, the speed of P is w and the speed of Q is $2w$.

The direction of motion of Q is reversed by the collision.

The impulse received by Q in the collision has magnitude $\frac{7}{2}mu$.

(a) Find w in terms of u .

(3)

(b) Find the two possible values of k .

(5)

Q24 **PAST PAPER** Pearson IAL WME01 · January 2026 · Q1**M1 · TOPIC 9** **6 marks**

1. A particle P of mass $3m$ and a particle Q of mass $5m$ are on a smooth horizontal surface. The particles move towards each other in opposite directions along the same straight line and collide.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $4u$

The magnitude of the impulse exerted on P by Q in the collision is $\frac{33}{2}mu$

Find

- (i) the speed of P immediately after the collision,
- (ii) the speed of Q immediately after the collision.

(6)

05 Momentum with SUVAT, Friction and Slopes

The hardest questions put a collision in the middle of a motion. Split the story into stages.

WHEN YOU SEE → YOU DO

“dropped from a height” / “bounces”

→ $v^2 = u^2 + 2as$ before and after the impulse

“rough”, “coefficient of friction μ ”

→ $F = \mu R$, then $F = ma$ gives the deceleration

“comes to rest after ... m / ... s”

→ SUVAT with $v = 0$

“inclined at θ ”, smooth slope

→ $a = g \sin \theta$ along the slope

WORKED MODEL • THE DROPPED BALL

0.16 kg dropped from 2 m; impulse on the ground 1.44 N s. Greatest height after?

1 $v^2 = 2(9.8)(2) \Rightarrow v = 6.26 \text{ m s}^{-1}$ down

stage 1: SUVAT

2 $1.44 = 0.16(w + 6.26) \Rightarrow w = 2.74 \text{ m s}^{-1}$ up

stage 2: impulse

3 $0 = 2.74^2 - 2(9.8)h$

stage 3: SUVAT

ANSWER $h = 0.383 \text{ m}$

IDEA 05

Practise: Momentum with SUVAT and Friction

Questions written to practise exactly this idea.

Q25**DR ESLAM**

Rough ground after a collision

M1 · TOPIC 9**4 marks**

After a collision a 3 kg block slides at 4 m s^{-1} on rough ground, $\mu = 0.2$. How far does it travel before it stops?

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|--|
| 00 | Always | Before/after diagram • positive direction • all “after” arrows positive. |
| 01 | One body | $I = mv - mu = Ft$. Rebound: the speeds add. Units N s. |
| 02 | Collision | $\sum mu = \sum mv$. The sign of the answer gives the direction. |
| 03 | Together | Coalesce / couple / string taut → one common velocity. |
| 04 | Impulse given | Use it on that particle; the other gets the opposite. “Two values” → $\pm$. |
| 05 | In a story | Before (SUVAT) → collision (momentum) → after (SUVAT). |

BOTTOM LINE Diagram → positive direction → conservation for the collision, impulse for one particle → read the direction from the sign.

Answers

Momentum, Impulse & Collisions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • MOMENTUM AND IMPULSE ON ONE BODY

- Q1** Dr Eslam • Momentum and units $15 \text{ m s}^{-1}, 18\,000 \text{ N s}$
- Q2** Dr Eslam • Rebound: the speeds add $0.2(4 + 6) = 2 \text{ N s}$
- Q3** Dr Eslam • Impulse = force $\times$ time $I = 12; 2(v - 5) = 12 \Rightarrow v = 11 \text{ m s}^{-1}$

IDEA 02 • COLLISIONS: CONSERVATION

- Q4** Dr Eslam • Draw, then conserve $24 - 6 = 4 + 2v \Rightarrow v = 7 \text{ m s}^{-1}$, in A's original direction
- Q5** Dr Eslam • Let the sign decide $10 = 2v + 12 \Rightarrow v = -1$: speed 1 m s^{-1} , reversed
- Q6** Pearson IAL WME01 • January 2022 • Q2 (a) $k = \frac{1}{3}$ (b) $1.5mu$
- Q7** Pearson IAL WME01 • January 2023 • Q2 (a)(i) 0.5 (ii) 1.5 m s^{-1} (b) $3m \text{ N s}$
- Q8** Pearson IAL WME01 • June 2023 • Q1 (a) $0.5u$ (b) reversed (c) $10u \text{ N s}$
- Q9** Pearson IAL WME01 • January 2024 • Q2 (a) $x = 2.5$ (b) 10 N s
- Q10** Pearson IAL WME01 • October 2024 • Q1 (a) $5x = 11y$ (b) $\frac{108}{11}mx$

IDEA 03 • ONE COMMON VELOCITY

- Q11** Dr Eslam · Coalesce $12 = 5v \Rightarrow v = 2.4 \text{ m s}^{-1}$
- Q12** Dr Eslam · String becomes taut $v = u; l = 2mu$
- Q13** Pearson IAL WME01 · October 2019 · Q1 (a) $1.2u$ (b) $8.4mu$
- Q14** Pearson IAL WME01 · October 2022 · Q1 (a) 1.6 m s^{-1} (b) $48\,000 \text{ N s}$
- Q15** Pearson IAL WME01 · June 2024 · Q1 (a) $S = \frac{U}{4}$ (b) $\frac{3}{4}mU$
- Q16** Pearson IAL WME01 · June 2026 · Q1 (a) $\frac{u}{16}$ (b) $\frac{75}{16}mu$

IDEA 04 · IMPULSE IN A COLLISION

- Q17** Dr Eslam · Use the impulse given $v = 8 \text{ or } v = 2 \text{ (both in the original direction)}$
- Q18** Pearson IAL WME01 · January 2020 · Q1 (a) $\frac{4}{3}m_2u$ (b) $\frac{(4m_2 - 3m_1)u}{3m_1}$ (c) speed > 0
- Q19** Pearson IAL WME01 · January 2021 · Q2 (a) $0.5u$ (b) $3u$
- Q20** Pearson IAL WME01 · June 2021 · Q1 (a) $25mu$ (b) $k = \frac{28}{3} \text{ or } \frac{22}{3}$
- Q21** Pearson IAL WME01 · June 2022 · Q1 (a) $\frac{2}{3}u$ (b) unchanged
- Q22** Pearson IAL WME01 · June 2025 · Q1 (a) $v = 3.5u$ (b) $k = 1.5$
- Q23** Pearson IAL WME01 · October 2025 · Q2 (a) $w = \frac{u}{12}$ (b) $k = \frac{7}{12} \text{ or } \frac{5}{12}$
- Q24** Pearson IAL WME01 · January 2026 · Q1 (i) $2.5u$ (ii) $0.7u$

IDEA 05 · MOMENTUM WITH SUVAT AND FRICTION

Q25 Dr Eslam · Rough ground after a collision

$$a = 0.2g = 1.96; \quad s = \frac{16}{2(1.96)} = 4.08 \text{ m}$$